

Aystro

Automated Retail Shelf Auditing: Technical Progress and Investment Readiness

A Report to Leadership and Investors

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Executive Summary

This report answers one question directly: is the computer-vision architecture Aystro is building on the right one to keep investing in, or should scope change? The short answer is that the direction is validated by measured evidence, not just design reasoning, and the evidence is now real rather than projected — trained models, real accuracy numbers, and a working application, not a proposal.

The core architectural bet — separating "where is a product" (detection) from "what product is it" (classification), instead of one model doing both — has been built and measured against the original single-stage approach. The new detector reaches higher accuracy and precision using roughly a third of the training images the original approach needed, at a training cost increase of well under one percent of a single month's cheapest paid Roboflow plan. Adding a new flavour variant to an existing brand — demonstrated in practice for Fanta during this engagement — took hours of labelling and one small classifier, not a retrain of the core system. That is the economic case for this architecture in one sentence: catalogue growth is now a data task, not an engineering one.

The report is equally direct about what is not yet proven. Classifier-layer accuracy has not yet been formally measured (a same-day fix). Two newer signals — Coca-Cola flavour variants and packaging material — are mid-experiment and constrained by real data gaps, not model performance: the available Coca-Cola photographs turned out on inspection to be almost entirely one variant, and only two genuine glass-packaging examples exist across the whole labelled corpus. Both gaps have a known, inexpensive fix — targeted photography — rather than an open-ended research problem. A geometric packaging-format rule described in earlier work has been suspended in production because it did not generalise reliably enough; the code is kept, not discarded, pending a better approach. These are stated plainly because a credible investment case rests on knowing the real gaps, not on a report with none.

1. The Business Problem

A retail brand's on-shelf presence — how many units are stocked, how they are arranged, and what share of a shelf each competitor occupies — is commercial information manufacturers and retailers pay to know, and today that information is gathered by a person walking the aisle with a clipboard or a phone. Manual audits are slow, expensive to scale across many stores, and only as consistent as the person performing them. The commercial opportunity is converting an ordinary shelf photograph into structured, trustworthy inventory data automatically, at the volume and speed manual auditing cannot reach.

The technical challenge is that this is not one problem but two, asked of every object in the photograph at once: where is it (localisation), and what is it (identity). A system that answers both questions with a single monolithic model inherits a serious commercial liability, examined next, that determines whether the product can keep pace with a retail catalogue that never stops changing.



Figure 1. A field capture with its detections overlaid; colour denotes brand. The density of a real audit scene — many products, tightly packed, photographed under ordinary store lighting — is the actual operating condition this system is built for, not a controlled benchmark.

2. What Was Built: The Architecture

A single model that classifies each product directly by identity must be retrained every time a new stock-keeping unit (SKU) enters the catalogue — expensive in annotation effort and compute, and risking regressions on products that were previously working. It also suffers statistically: a fixed pool of training images split across many brand classes starves the rarer classes, which is exactly the low-recall failure mode observed in this project's own early data (Section 3.1).

The system built during this engagement separates the volatile part of the problem (product identity, which changes constantly) from the stable part (object presence, which does not), and narrows the decision space one stage at a time:

Stage 1 — Detection. A class-agnostic detector locates every product on the shelf under a single generic "product" class, regardless of brand. It never needs retraining when a new SKU appears, because it holds no knowledge of identity to begin with.

Stage 2 — Brand classification. Each detected region is cropped and classified by manufacturer brand. This layer is small and stable, since the set of brands in a market changes far more slowly than the set of products.

Stage 3 — Variant classification. Within a brand, a further classifier resolves the specific flavour or product variant. Built and shipped for Fanta during this engagement; Coca-Cola is in progress (Section 3.2).

An experimental fourth signal — packaging material (glass / plastic / can), independent of brand — is under evaluation in parallel (Section 3.3).

This is implemented as a live, hierarchical Roboflow Workflow, not a proposal: a shared detector feeds crops to a brand classifier and a flavour classifier running in parallel, and a small merge step selects the flavour label only when the brand it belongs to actually has one, otherwise passing the plain brand through unchanged. Restricting brand-variant comparison to one manufacturer's catalogue rather than the entire market also reduces visual confusion between candidates.

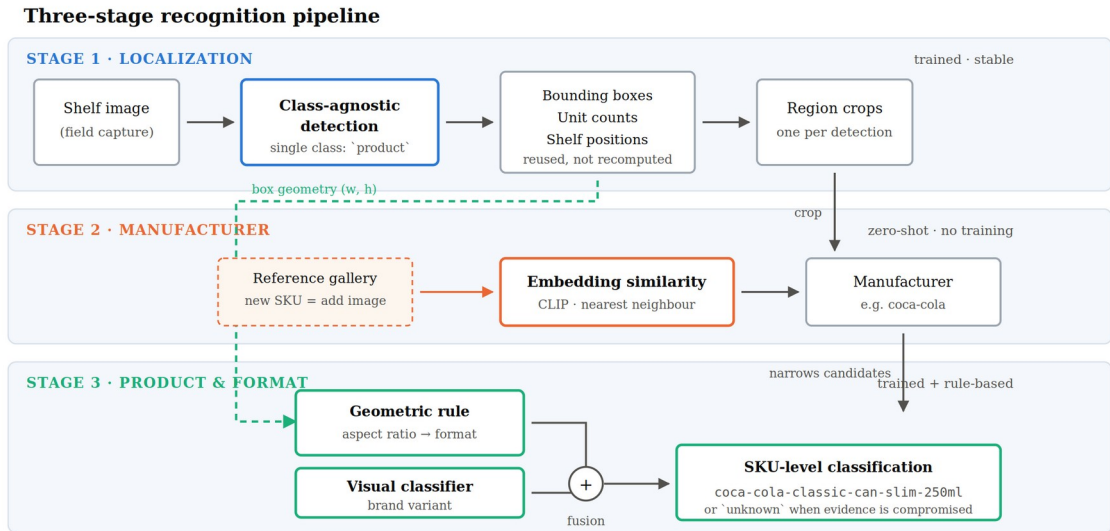


Figure 2. Block diagram of the staged pipeline: class-agnostic detection, region cropping, brand classification, and variant classification, with packaging material as an independent, parallel signal rather than a further narrowing of the same decision.

3. Evidence: Does It Work?

3.1 Detector accuracy: architecture change, measured

The original brand-classed detector and the current class-agnostic detector were compared at their most mature, most fairly matched versions — both trained on the same model architecture (RF-DETR-large) — rather than against an early, undertrained snapshot of either.

Table 1. Detector accuracy, original brand-classed architecture versus the current class-agnostic architecture, both on the RF-DETR-large model.

Measure	Original (brand-classed, 6 classes)	Current (class-agnostic)
Training images	1,191	384
mAP50	86.92%	89.3%
Precision	82.7%	90.5%
Recall	85.8%	81.1%

The current architecture reaches higher precision and overall accuracy using roughly a third of the training images the original approach needed — a genuine data-efficiency result, not merely a design argument. Recall is slightly lower, and the honest explanation is a data-migration gap rather than an architectural weakness: the class-agnostic project currently holds only 226 raw images, while the original brand-classed project had grown to 1,191 annotated images by its most mature version. Roughly 800 already-labelled images have not yet been carried over.

Migrating them is inexpensive — a programmatic relabel to a single "product" class, not new annotation work — and is the highest-value near-term action identified in this report (Section 7).

3.2 Classifier accuracy: a known, closeable gap

The brand classifier (Stage 2) and the Fanta flavour classifier (Stage 3) are both trained and serving live traffic, but neither has a formally recorded accuracy evaluation yet — the training platform tracks this separately from the training run itself, and that follow-up step has not been run. This report states that plainly rather than substituting an estimate: no classifier accuracy figure is claimed here that has not been measured. Running the missing evaluation is a same-day task and the first item on the roadmap in Section 7.

A second classifier, for Coca-Cola flavour variants, was built and seeded with the available labelled photography, then audited by hand before training. That audit found the available Coca-Cola images were almost entirely one variant ("classic") with negligible real diversity — so rather than train a flavour classifier the data cannot actually support, the project was left as a single class pending genuinely varied new photography. This is a data-collection gap, not a modelling one, and is called out explicitly in Section 6.

3.3 Packaging-material classifier: an experiment in progress

A new, independent signal — packaging material (glass, plastic, or can) — is being evaluated separately from brand and flavour, on the basis that material is brand-independent: a plastic bottle looks plastic regardless of manufacturer, so one shared classifier serves every brand and every future brand added to the catalogue, rather than needing a per-brand model.

A full visual audit of the 1,608 images available across all six brands found only two genuine glass-packaging examples in the entire corpus — far too few to train a reliable glass class. Rather than force those two images into a mislabelled bucket or overstate a third category, the experiment was scoped down to a two-class problem, plastic versus can, with a workable, roughly balanced split of 860 plastic and 746 can images. Image upload for this experiment is complete; labelling and training are in progress at the time of writing. This is reported as an open, honest in-progress item, not a finished result.

3.4 Engineering maturity: caught, fixed, and verified against live data

Two incidents from this engagement are included deliberately, as evidence of process rather than as embarrassments to omit. First, a defect in the workflow's custom merge logic was silently turning full class names into single stray characters in the live application; it was diagnosed from a user report, root-caused to a data-shape assumption that training-time metrics could not have caught, fixed, and verified against a real photograph within the same working session. Second, a

geometric rule for distinguishing packaging formats by box shape — a promising result on the specific example it was measured against — did not generalise reliably enough in broader use and has been suspended in production while a more accurate approach is developed; the code and its tests are kept intact, not deleted, for that follow-up. Both are the kind of finding a system only surfaces once it is actually run against real, uncontrolled data — which is itself part of the evidence that this project has moved past the design stage.

4. Economics: The Cost to Scale

The central commercial risk in any catalogue-driven product is that growth becomes expensive: every new SKU, every new market, every new customer could in principle demand new engineering. This architecture was specifically chosen to avoid that, and the claim is backed by this engagement's own measured costs rather than a projection.

Table 2. Actual training cost of every model in the current production stack, at Roboflow's published rate of one credit per thirty minutes of GPU training.

Model	Training time	Training credits
Original detector (first version)	11.2 min	0.37
Current production detector	44.2 min	1.48
Brand classifier (Stage 2)	9.1 min	0.30
Fanta flavour classifier (Stage 3)	7.2 min	0.24

The entire current three-model production stack has cost roughly 2 credits (about one hour of GPU time) to train, in total, ever — a small fraction of a single month's allowance on even the cheapest paid plan. Moving from one model to three did not meaningfully raise training cost while measurably raising accuracy (Section 3.1). Current dollar pricing changes independently of this report and should always be checked directly against Roboflow's published rates rather than quoted here as a fixed figure.

The more consequential number for an investor is the marginal cost of catalogue growth, and this engagement produced a direct, not hypothetical, data point: adding Fanta flavour differentiation — a new axis of product identity — required labelling a few hundred already-available product images and training one small classifier, wired into the pipeline as a parallel step. No change to the detector or the brand classifier was needed. It was completed within hours, not weeks. The same pattern applies to onboarding an entirely new company: the backing database is multi-tenant by design, so a new customer is a data-entry operation, not a code change.

One cost is real and not yet measured, and is stated here rather than left out: the current pipeline issues a brand-classification and a flavour-classification call for every detected product in a

photograph, not once per photograph, so a busy shelf with twenty products now triggers meaningfully more classification calls than the original single-model approach. This should be measured directly from live workflow usage before it is quoted as a number, and is flagged as an open item in Section 6.

5. The Application: Field-Proven

A cross-platform application, developed alongside the model architecture, is what turns model output into a usable business process rather than a research demo. It standardises field capture, runs the detection pipeline against each photograph, and — critically — interposes a human reviewer between raw model output and any figure that gets reported as fact.

Since the start of this engagement the application has grown past initial capture-and-review into reporting the business actually asked for: a Submissions view so a company administrator can see what a field representative actually recorded (previously, captured data was written to the database but nothing in the application read it back), and a brand-to-variant Share-of-Shelf report giving each brand's overall percentage of shelf space broken down into the specific variants the pipeline currently distinguishes. A low-quality capture warning — a photograph shot too far from the shelf to count reliably — was also upgraded from an easy-to-miss notification to a dialog the representative must actively dismiss, directly reducing a source of silent undercounting in the field.



Figure 3. The application reviewing a live capture, with detections and a share-of-shelf summary computed beneath — the point at which an automated count becomes a number a person has actually verified before it is reported.

6. Risks and Open Gaps

Stated in the order they should be closed, with the honest reason each remains open:

1. Classifier accuracy is unmeasured. The brand and Fanta-flavour classifiers are live but have no recorded evaluation. Closing this is a same-day task with no dependency on new data.
2. The detector's recall gap traces to a specific, known cause. Roughly 800 already-annotated images from the original project have not been migrated into the class-agnostic one. This is a data-engineering task, not new research.
3. The packaging-material classifier is untrained and, more fundamentally, has almost no glass examples to learn from — two images across the entire labelled corpus. Plastic-versus-can is

workable with existing data; a genuine glass class requires targeted new photography, not more labelling of what already exists.

4. Coca-Cola flavour variety does not exist in the current data. The available photographs are almost entirely one variant. This is the same fix as item 3 — new photography of the missing variants — and should not be attempted by relabelling existing images, which the audit already ruled out.

5. Per-product inference cost at scale has not been measured. The architecture change moved from one classification call per photograph to several per detected product; the real cost impact should be measured from live usage before it is used in a cost projection.

7. Recommendation and Next Steps

The recommendation is to continue the current architecture. The evidence in this report is that it is not a trade-off between cost and accuracy against the original approach — it is measurably more accurate on a third of the training data, at a training-cost increase too small to matter, and it has already demonstrated in practice, not in theory, that catalogue growth is cheap: the Fanta flavour layer was added in hours. The open items in Section 6 are specific, individually inexpensive, and none of them point to a flaw in the underlying design.

Proposed order of work: first, run the missing classifier evaluation and migrate the roughly 800 unmigrated detector images, retraining the detector against the fuller dataset — both are days of work with no new data collection required and directly close Sections 6.1 and 6.2. Second, decide whether to fund a short, targeted photography pass covering genuine Coca-Cola flavour variants and glass-packaged products specifically; without it, those two experiments cannot progress further regardless of engineering effort, since the constraint is data that does not yet exist, not model capability. Third, measure real per-photograph inference cost from live usage once volume justifies it, to keep the economics in Section 4 current rather than assumed.

What this report asks of leadership is a decision to continue funding the current direction on the strength of measured results rather than a promising design. What it offers an investor is a concrete answer to "does the core technology work": yes, measurably, at a training cost too small to be a meaningful line item, with a demonstrated, not projected, path to adding new products cheaply — and a team that finds its own defects, states its own gaps, and fixes rather than hides them.

Technical References

- [1] Radford, A., Kim, J. W., Hallacy, C., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. Proceedings of the 38th International Conference on Machine Learning (ICML). — background for the embedding-based classification approach used in Stages 2 and 3.
- [2] Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Advances in Neural Information Processing Systems (NeurIPS). — background for the detect-then-classify architectural pattern underlying Stage 1.
- [3] Roboflow. Workflows Documentation. <https://inference.roboflow.com/workflows/> (accessed August 2026). — the platform the production pipeline in Section 2 is implemented on.
- [4] Roboflow. Pricing and Credits. <https://roboflow.com/pricing> (accessed August 2026). — source for the credit rates used in Section 4; consult directly for current dollar pricing.